22 There are three buses, **A**, **B** and **C**.

The total number of seats in the three buses is 250

28% of the 250 seats are in bus **A**.

number of seats in bus **B** : number of seats in bus **C** = 3 : 2

38 of the seats in bus **B** are empty.
26 of the seats in bus **C** are empty.

Jamie says,

> "The percentage of the seats in bus **B** that are empty is greater than the percentage of the seats in bus **C** that are empty."

Is Jamie correct?
You must show how you get your answer.

Bus A: 250 × 0.28 = 70 seats

1 mark
Find the number of seats in A.

B and C: 250 − 70 = 180 seats

1 mark
Share only the remaining seats.

One part = 180 ÷ (3 + 2) = 36

Bus B: 3 × 36 = 108; bus C: 2 × 36 = 72

1 mark
Divide in the ratio 3 : 2.

B empty: 38 ÷ 108 × 100 = 35.185...%

C empty: 26 ÷ 72 × 100 = 36.111...%

1 mark
Use the capacity of EACH bus as its denominator.

No. 35.185...% < 36.111...%, so C has the greater percentage of empty seats.

1 mark
Compare percentages, not the empty-seat counts.

(Total for Question 22 is 5 marks)

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